

DNSGuard AI

Behavioral DNS threat detection with Splunk MLTK

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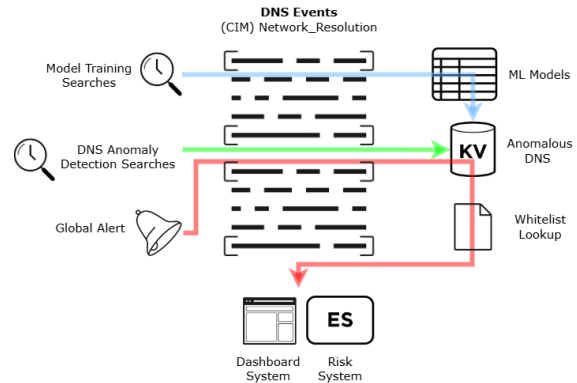
1 Problem Statement

The **Domain Name System (DNS)** is a fundamental pillar of modern network communication, responsible for resolving hostnames to IP addresses. Despite its critical role, **DNS** often remains **under-monitored** within enterprise security strategies, making it an attractive vector for threat actors. Advanced adversaries increasingly leverage DNS to establish **command-and-control (C2)** channels, exfiltrate **sensitive information**, and perform **stealth reconnaissance activities**—all while blending within legitimate traffic patterns. These attacks frequently exploit the inherent trust and ubiquity of **DNS**, using low-volume, protocol-compliant queries that bypass signature-based intrusion detection systems. The subtlety and evasiveness of **DNS-based threats** demand a behavioral and data-driven approach to detection.

DNSGuard AI addresses this security gap by applying unsupervised machine learning and statistical analysis to **DNS telemetry** within the Splunk platform. By modeling baseline DNS behavior and detecting deviations indicative of malicious activity, **DNSGuard AI** provides real-time visibility into anomalous patterns that traditional tools fail to uncover—offering a scalable and adaptive defense mechanism for enterprise environments.

2 Solution Overview

The architecture illustrates how **DNSGuard AI** processes DNS events mapped to the **Network_Resolution** data model in Splunk. **Model training** searches extract features from historical DNS traffic to train machine learning models via the **MLTK**. In parallel, **anomaly detection** searches continuously scan incoming DNS data to identify suspicious behavior such as **exfiltration**, **tunneling**, or **domain shadowing**. Detected anomalies are stored in a **KV Store collection** and compared against a **whitelist** to suppress false positives. Validated anomalies are then sent to two systems: the **dashboard interface** for visual monitoring, and **Splunk Enterprise Security (ES)** for risk scoring and alert generation. This design ensures scalable, real-time DNS threat detection tightly integrated with Splunk’s security ecosystem.



3 Machine Learning Approach

The detection engine of **DNSGuard AI** integrates multiple unsupervised learning algorithms to identify deviations from normal DNS behavior across diverse threat dimensions:

- **Density Function Algorithm**

- **Beaconing:** Detects regular, periodic DNS queries at consistent intervals—a hallmark of malware communicating with command and control servers. Analyzes consistency of time gaps between queries to the same domain

- **Anomaly Detection Algorithm**

- **Record Type Anomalies:** Detects abnormal usage of specific DNS record types often associated with reconnaissance or data exfiltration. Identifies anomalies in the usage of TXT (data exfiltration), ANY (broad queries), HINFO (host info leakage), and AXFR (zone transfer attempts) records by host.
- **C2 Tunneling Detection:** Identifies hosts making an unusually high number of DNS queries, which could indicate command and control communication or data exfiltration through DNS tunneling.
- **Query Length Anomalies:** Detects unusually long DNS queries that may represent data exfiltration channels where sensitive information is encoded in the query itself. Identifies anomalies in query string length by host.
- **Domain Shadowing:** Identifies patterns where many unique subdomains are requested for a legitimate domain, which may indicate an attacker using compromised DNS accounts to create malicious subdomains. Measures distinct subdomain count by parent domain and identifies anomalies.

- **K Means Algorithm**

- **Behavioral Clustering:** Groups hosts with similar abnormal DNS behavior, which can reveal coordinated attacks or infected host groups across the enterprise. Uses KMeans clustering on multiple DNS behavior features.

4 Installation and Configuration

Prerequisites

Install the following from Splunkbase:

- Splunk Enterprise/Cloud 8.0+
- Splunk CIM App
- Splunk MLTK App
- Python for Scientific Computing

1. Install the App

- Clone the repo:

```
git clone https://github.com/aleeric/Splunk-DNSGuard-AI.git
```

- Move the folder to `$SPLUNK_HOME/etc/apps/` and restart Splunk.

2. Generate Test Data

```
cd Synthetic-Data
python generate_dns_events.py
```

3. Import Data into Splunk

Via Web Interface:

- Go to **Settings** → **Add Data** → **Upload**
- Select `dns_events.json`, set **Source type** to `_json`, create index `synthetic_data`
- Click **Next** → **Submit**

4. Map to Network_Resolution Model

- Go to **Settings** → **Data Models** → **Network_Resolution**
- Click **Edit** → **Add Dataset**, include `synthetic_data`, then save

5. Verify Setup

- Open the app in Splunk Web
- Check dashboards and validate that detections work

5 Conclusion

DNSGuard AI gives security teams an advanced and scalable solution to detect DNS threats in real time. It combines machine learning, Splunk's search power, and a modular dashboard system to identify C2, exfiltration, and reconnaissance via DNS, filling a critical detection gap in enterprise defense.